

Day 32 (Part 2): IPv6

IPv6 ADDRESS CONFIGURATION (EUI-64)

- EUI stands for Extended Unique Identifier
- (Modified) EUI-64 is a method of converting a MAC address (48-bits) into a 64-bit INTERFACE identifier
- This INTERFACE identifier can then become the "HOST portion" of a /64 IPv6 ADDRESS

How to convert the MAC address:

1: Divide the MAC address in half

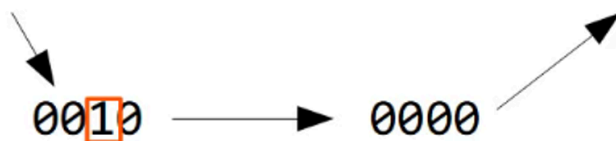
1234 5678 90AB → 1234 56 | 78 90AB

2: Insert FFFE in the middle

1234 56FF FE78 90AB

3: Invert the 7th bit

1234 56FF FE78 90AB → 1034 56FF FE78 90AB



EUI-64 PRACTICE:

```
782B CBAC 0867 >>> 782B CB || AC 0867
```

```
782B CBFF FEAC 0867
```

8 is the 7th bit so 1000 inverted becomes 1010 = A in hex

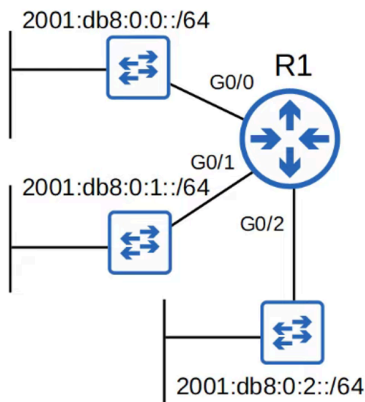
so the EUI-64 Interface Identifier is : 7A2B CBFF FEAC 0867

Configuring IPv6 addresses (EUI-64)

MAC Address	EUI-64 Interface Identifier
782B CBAC 0867	7A2B CBFF FEAC 0867
0200 4C4F 4F50	0000 4CFF FE4F 4F50
0050 56C0 0001	0250 56FF FEC0 0001
00FF 6BA6 F456	02FF 6BFF FEA6 F456
96AB 6D6B 98AE	94AB 6DFF FE6B 98AE

CONFIGURING IPv6 ADDRESSES with EUI-64

Configuring IPv6 addresses (EUI-64)



```

R1(config)#int g0/0
R1(config-if)#ipv6 address 2001:db8::/64 eui-64
R1(config-if)#no shutdown
R1(config-if)#
R1(config-if)#int g0/1
R1(config-if)#ipv6 address 2001:db8:0:1::/64 eui-64
R1(config-if)#no shutdown
R1(config-if)#
R1(config-if)#int g0/2
R1(config-if)#ipv6 address 2001:db8:0:2::/64 eui-64
R1(config-if)#no shutdown
  
```

```
R1#show interfaces g0/0
GigabitEthernet0/0 is administratively down, line protocol is down
Hardware is iGbE, address is 0cf8.2236.8500 (bia 0cf8.2236.8500)
```

```
R1#show interfaces g0/1
GigabitEthernet0/1 is administratively down, line protocol is down
Hardware is iGbE, address is 0cf8.2236.8501 (bia 0cf8.2236.8501)
```

```
R1#show interfaces g0/2
GigabitEthernet0/2 is administratively down, line protocol is down
Hardware is iGbE, address is 0cf8.2236.8502 (bia 0cf8.2236.8502)
```

```
R1(config-if)#do show ipv6 interface brief
GigabitEthernet0/0    [up/up]
FE80::EF8:22FF:FE36:8500
2001:DB8::EF8:22FF:FE36:8500
GigabitEthernet0/1    [up/up]
FE80::EF8:22FF:FE36:8501
2001:DB8:0:1:EF8:22FF:FE36:8501
GigabitEthernet0/2    [up/up]
FE80::EF8:22FF:FE36:8502
2001:DB8:0:2:EF8:22FF:FE36:8502
GigabitEthernet0/3    [administratively down/down]
unassigned
```

NOTE the "2001:DB8..." Address has "E" changed to "c". This is the 7th bit getting flipped (1110 to 1100 = 12 = hex 'C')

WHY INVERT THE 7th BIT ?

- MAC addresses can be divided into TWO TYPES:
 - UAA (Universally Administered Address)
 - Uniquely assigned to the device of the manufacturer
 - LAA (Locally Administered Address)
 - Manually assigned by an Admin (with the mac-address command on the INTERFACE) or protocol. Doesn't have to be globally unique.

- You can IDENTIFY a UAA or LAA by the 7th bit of the MAC ADDRESS, called the U/L bit (Universal/Local bit)
 - U/L bit set to 0 = UAA
 - U/L bit set to 1 = LAA
- In the context of IPv6 addresses/EUI-64, the meaning of the U/L bit is reversed:
 - U/L bit set to 0 = The MAC address the EUI-64 INTERFACE ID was made from was an LAA
 - U/L bit set to 1 = The MAC address the EUI-64 INTERFACE ID was made from was a UAA

IPv6 ADDRESS TYPES

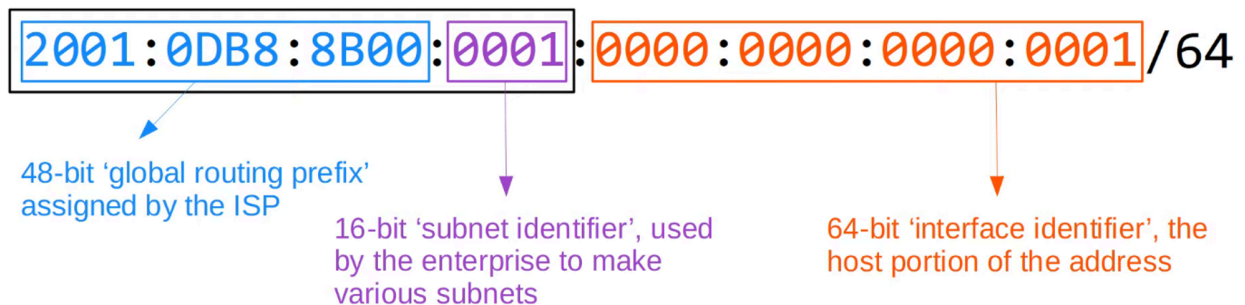
1. GLOBAL UNICAST ADDRESSES

- **Global Unicast** IPv6 ADDRESSES are PUBLIC ADDRESSES which can be used over the INTERNET
- Must REGISTER to use them.
- They are PUBLIC ADDRESSES so need to be GLOBALLY UNIQUE

💡 Originally defined as the 2000 :: /3 block (2000:: to 3FFF : FFFF : FFFF : FFFF : FFFF : FFFF : FFFF : FFFF)

- NOW defined as ALL ADDRESSES which are not RESERVED for other purposes

Remember THESE THREE PARTS of a GLOBAL UNICAST ADDRESS



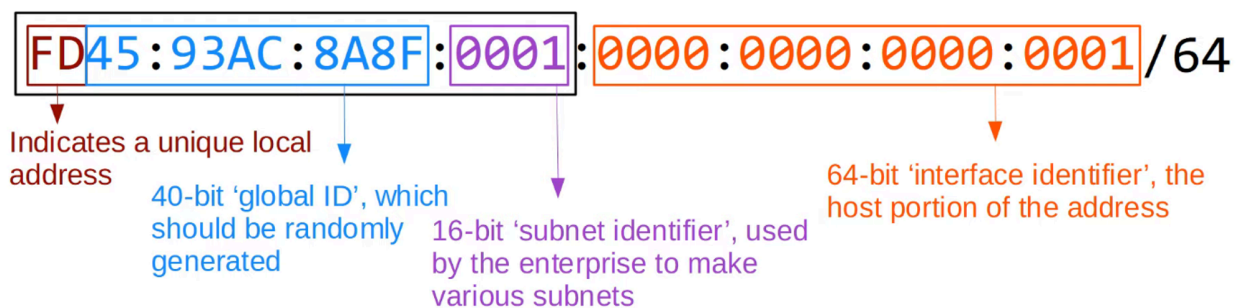
1. UNIQUE LOCAL ADDRESSES

- **Unique Local** IPv6 ADDRESSES are PRIVATE ADDRESSES which cannot be used over the internet
- You do NOT need to REGISTER to use them
- Can be used FREELY within INTERNAL NETWORKS
- Do NOT need to be GLOBALLY UNIQUE (*)
- CANNOT be ROUTED over the INTERNET

💡 Uses the ADDRESS block FC00 ::/7 (FC00:: to FDFF : FFFF : FFFF : FFFF : FFFF : FFFF : FFFF)

- A later UPDATE required the 8th bit to be set to 1 so the FIRST TWO DIGITS must be FD

(*) The GLOBAL ID should be UNIQUE so that ADDRESSES don't overlap when companies MERGE



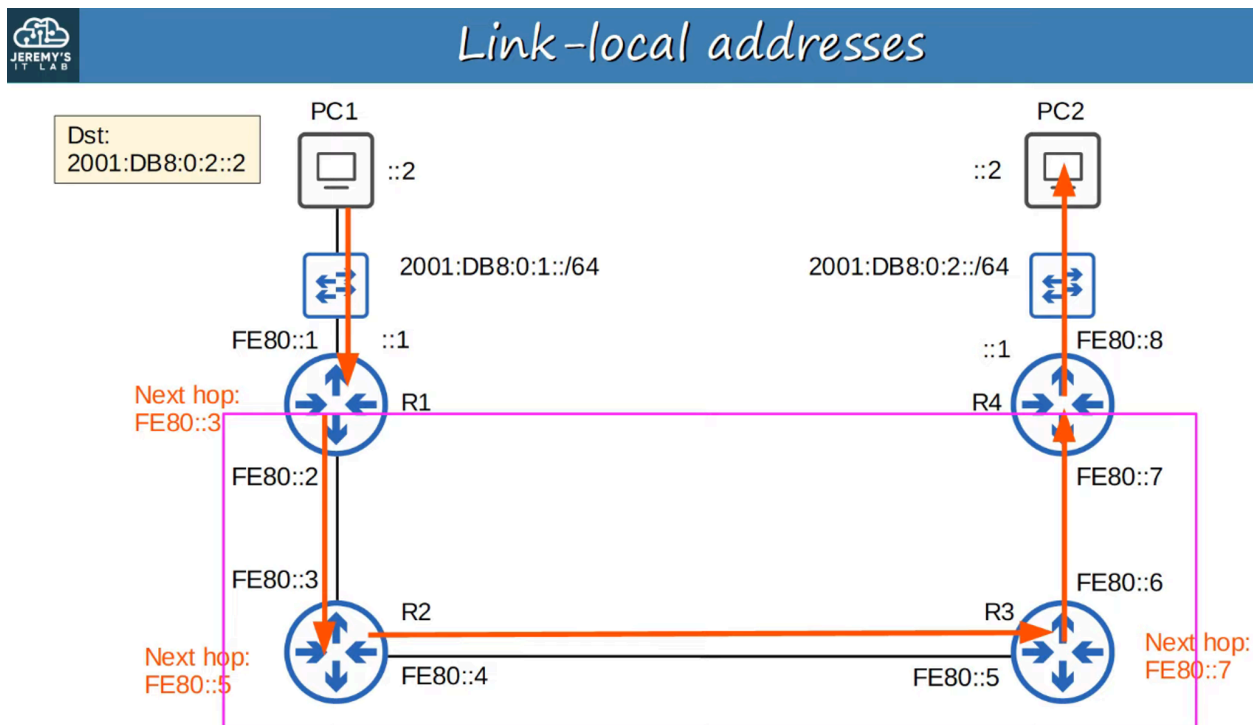
1. LINK-LOCAL ADDRESSES

- **Link-Local** IPv6 ADDRESSES are AUTOMATICALLY generated on IPv6-enabled INTERFACES
- Use command `R1(config-if)# ipv6 enable` on an interface to enable IPv6 on an INTERFACE

💡 Uses the ADDRESS block FE80::/10 (FE80:: to FEBF : FFFF : FFFF : FFFF : FFFF : FFFF : FFFF)

- The STANDARD states that the 54-bits AFTER FE80/10 should be ALL 0's so you won't see Link-Local ADDRESSES beginning with FE9, FEA, or FEB - ONLY FE8(!)
- The INTERFACE ID is generated using EUI-64 rules
- Link-Local means that these addresses are used for communication within a single link (SUBNET)
 - ROUTER will not route PACKETS with a Link-Local DESTINATION IPv6 ADDRESS
- Common uses of Link-Local Addresses:
 - Routing Protocol Peerings (OSPFv3 uses Link-Local Addresses for Neighbour Adjacencies)
 - NEXT-HOP ADDRESS for STATIC ROUTES
 - Neighbor Discovery Protocol (NDP, IPv6's replacement for ARP) uses Link-Local ADDRESSES to function

Network using Link-Local Addresses for "next-hop" routing



1. MULTICAST ADDRESSES

- **Unicast Addresses** are one-to-one
 - ONE SOURCE to ONE DESTINATION
- **Broadcast** Addresses are one-to-all
 - ONE SOURCE to ALL DESTINATIONS (within the subnet)
- **Multicast** Addresses are one-to-many
 - ONE SOURCE to MULTIPLE DESTINATIONS (that have joined the specific *multicast* group)

💡 IPv6 uses range FF00::/8 for multicast (FF00:: to FFFF : FFFF : FFFF : FFFF : FFFF : FFFF : FFFF : FFFF)

- **IPv6 doesn't use Broadcast** (there IS NO "Broadcast Address" in IPv6!)

YOU MUST KNOW THE MULTICAST ADDRESS FOR EACH ROUTER TYPE

NOTE that the IPv6 and IPv4 Addresses share the same last digit

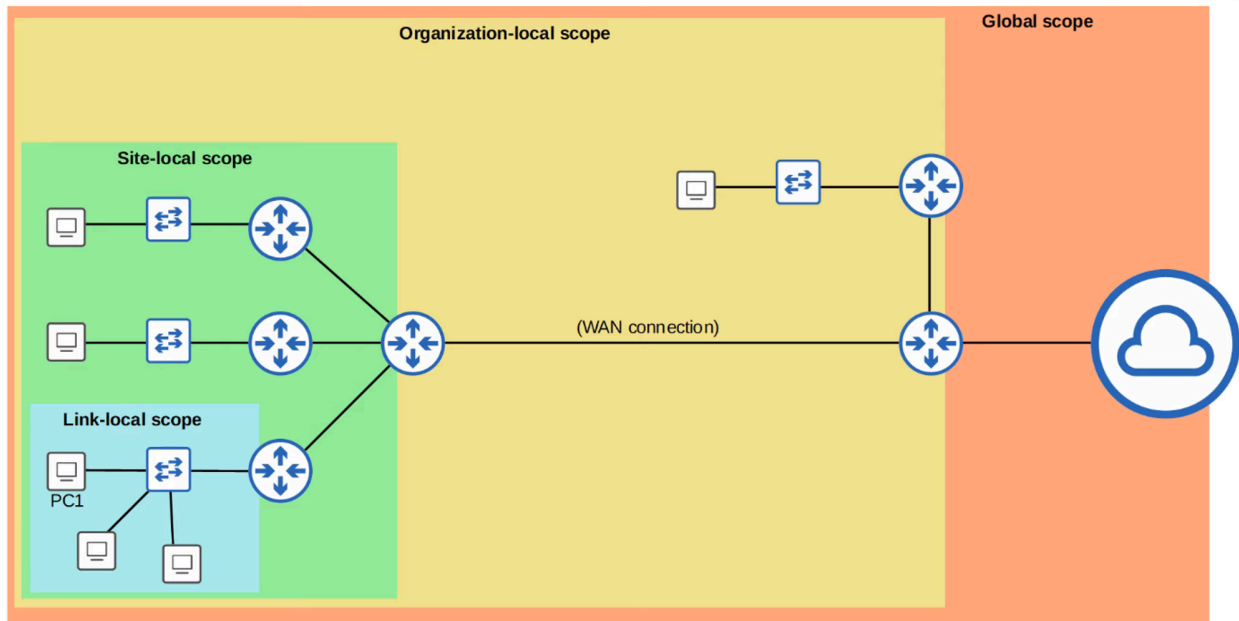
 JEREMY'S IT LAB	Multicast addresses	
Purpose	IPv6 Address	IPv4 Address
All nodes/hosts (functions like broadcast)	FF02::1	224.0.0.1
All routers	FF02::2	224.0.0.2
All OSPF routers	FF02::5	224.0.0.5
All OSPF DRs/BDRs	FF02::6	224.0.0.6
All RIP routers	FF02::9	224.0.0.9
All EIGRP routers	FF02::A	224.0.0.10

MULTICAST ADDRESS SCOPES

- IPv6 defines multiple MULTICAST 'scopes' which indicate how far the PACKET should be forwarded
- The ADDRESS in the previous slide all use the 'link-local' scope (FF02), which stays in the LOCAL SUBNET

IPv6 Multicast Scope Types:

- **Interface-Local (FF01)**
 - The PACKET doesn't leave the LOCAL device
 - Can be used to SEND traffic to a SERVICE within the LOCAL device
- **Link-Local (FF02)**
 - The PACKET remains in the LOCAL SUBNET
 - ROUTERS will not route the PACKET between SUBNETS
- **Site-Local (FF05)**
 - The PACKET can be forwarded by ROUTERS
 - Should be limited to a SINGLE PHYSICAL LOCATION (not forwarded over a WAN)
- **Organization-Local (FF08)**
 - Wider in scope than Site-Local (an entire company / ORGANIZATION)
- **Global (FF0E)**
 - No boundaries
 - Possible to be ROUTED over the INTERNET



1. ANYCAST ADDRESS

- **ANYCAST is a NEW feature of IPv6**
- ANYCAST is 'one-to-one-of-many'
- Multiple ROUTERS are configured with the SAME IPv6 ADDRESS
 - They use a ROUTING PROTOCOL to advertise the address
 - When HOSTS sends PACKETS to that DESTINATION ADDRESS, ROUTERS will forward it to the NEAREST ROUTER configured with THAT IP ADDRESS (based on ROUTING METRIC)
- There is NO SPECIFIC ADDRESS range for ANYCAST ADDRESSES.
 - Use a regular UNICAST (Global Unicast, Unique Local) and specify THAT as an ANYCAST ADDRESS
 - `R1(config-if)# ipv6 address 2000:db8:1:1::99/128 anycast`

Anycast address configuration

```
R1(config)#int g0/0
R1(config-if)#ipv6 address 2001:db8:1:1::99/128 anycast
R1(config-if)#
R1(config-if)#do show ipv6 interface g0/0
GigabitEthernet0/0 is up, line protocol is up
  IPv6 is enabled, link-local address is FE80::EF8:22FF:FE36:8500
  No Virtual link-local address(es):
  Global unicast address(es):
    2001:DB8::EF8:22FF:FE36:8500, subnet is 2001:DB8::/64 [EUI]
    2001:DB8:1:1::99, subnet is 2001:DB8:1:1::99/128 [ANY]
  Joined group address(es):
    FF02::1
    FF02::2
    FF02::1:FF00:99
    FF02::1:FF36:8500
  MTU is 1500 bytes
  ICMP error messages limited to one every 100 milliseconds
  ICMP redirects are enabled
  ICMP unreachables are sent
  ND DAD is enabled, number of DAD attempts: 1
  ND reachable time is 30000 milliseconds (using 30000)
  ND advertised reachable time is 0 (unspecified)
```

1. OTHER IPv6 ADDRESSES

- The :: Address = The *unspecified* IPv6 ADDRESS
 - Can be used when a DEVICE doesn't yet know its IPv6 ADDRESS
 - IPv6 DEFAULT ROUTES are configured to ::/0
 - IPv4 equivalent: 0.0.0.0
- The ::1 Address = The Loopback Address
 - Used to test the PROTOCOL STACK on the LOCAL DEVICE
 - Messages sent to THIS ADDRESS are processed within the LOCAL DEVICE but not SENT to other DEVICES
 - IPv4 equivalent : 127.0.0.0 /8 address range